Determinant–Zero Compatibility at the High-β Endpoint:
Exact Exponent Cone, Reserve Accounting, and Stop Certificates

Liang Wang

July 2026

Abstract

For the same literal fixed-h_0 post-bin coefficient, suppose one has separately proved

$$D_X \geq X^{-\lambda_D+o(1)} Q_X^3 J_X^2, \qquad |Z_X| \leq X^{-\eta_Z+o(1)} Q_X^2.$$

At the high-β scales

$$Q_X = X^{267/400+o(1)}, \qquad J_X = X^{133/400+o(1)},$$

we recover the exact flatness ledger

$$\frac{|Z_X|^2}{D_X} \leq X^{1/400-(2\eta_Z-\lambda_D)+o(1)}.$$

Hence the recorded pair reaches the TPC-32 threshold exactly in the closed exponent cone $\lambda_D \leq 2\eta_Z$. Equality is compatible but has zero determinant reserve.

We prove that this cone is logically independent of the strict physical synthesis condition $\Lambda_{\text{phys}} < 1/400$. A direct-product countermodel shows that no determinant reserve can, without a new intertwining theorem, be spent as physical condition-number slack. Equality in the physical ledger is therefore a stop even though equality in the determinant cone is allowed. Applying the audit to TPC-109–111, we find no certified exponent pair: the preceding papers prove exact saturation, kernel, and signed-prefix criteria, but neither a growing lower bound for D_X nor a growing upper bound for Z_X. The compatibility route remains open rather than passed or globally disproved.

Keywords: determinant energy; distinguished zero mode; exponent cone; reserve accounting; fixed shift; endpoint audit.

1 Typed inputs

Fix a prescribed $h_0 \neq 0$. The determinant statistics are

$$Z_X = \sum_n A_X(n), \qquad D_X = \sum_n |A_X(n)|^2.$$

Both must be formed after literal aggregation on the same actual support and under one normalization [6, 7]. Define

$$\mathfrak{F}_{h_0}(X) = \begin{cases} |Z_X|^2/D_X, & D_X > 0, \\ 0, & D_X = 0. \end{cases}$$

The zero branch is consistent because $D_X = 0$ forces $A_X = 0$ and hence $Z_X = 0$.

Definition 1.1 (Typed exponent certificate)**.** A pair $(\lambda_D, \eta_Z) \in [0, \infty)^2$ is *certified* only if the two displayed estimates

$$D_X \geq X^{-\lambda_D + o(1)} Q_X^3 J_X^2, \qquad |Z_X| \leq X^{-\eta_Z + o(1)} Q_X^2$$

are both proved for the same literal growing coefficient, the same fixed h_0, and the same quantifier range.

An upper bound for pre-bin energy cannot fill the first slot. A finite saturation theorem or a diagnostic ratio cannot fill either slot. An averaged-shift zero-mode theorem cannot fill the second slot without a proved return to the prescribed h_0.

2 The exact exponent cone

Theorem 2.1 (General flatness transfer)**.** *Let*

$$Q_X = X^{\beta + o(1)}, \qquad J_X = X^{1-\beta+o(1)}$$

with $0 < \beta < 1$*. Every certified pair satisfies*

$$\boxed{\mathfrak{F}_{h_0}(X) \leq X^{3\beta - 2 + \lambda_D - 2\eta_Z + o(1)}.}$$

Proof. Divide the square of the zero-mode upper bound by the energy lower bound:

$$\mathfrak{F}_{h_0}(X) \leq X^{\lambda_D - 2\eta_Z + o(1)} \frac{Q_X}{J_X^2}.$$

The last ratio is $X^{3\beta-2+o(1)}$. □

Corollary 2.2 (Critical exponent cone)**.** *At*

$$\beta = \frac{267}{400}, \qquad 1 - \beta = \frac{133}{400},$$

put

$$\rho_{\det} = 2\eta_Z - \lambda_D.$$

Then

$$\boxed{\mathfrak{F}_{h_0}(X) \leq X^{1/400 - \rho_{\det} + o(1)}.}$$

The two recorded estimates imply the TPC-32 threshold [5]

$$\mathfrak{F}_{h_0}(X) \leq X^{1/400 + o(1)}$$

exactly when

$$\boxed{\lambda_D \leq 2\eta_Z.}$$

Proof. The scale identity is

$$3\frac{267}{400} - 2 = \frac{1}{400}.$$

Now apply theorem 2.1. □

Remark 2.3 (Sharp scope). The word "exactly" concerns logical deduction from the two recorded bounds. If $\lambda_D > 2\eta_Z$, those bounds do not certify the threshold; the actual coefficient might satisfy sharper estimates.

3 Boundary and reserve classifications

Definition 3.1 (Determinant reserve)**.** The number

$$\rho_{\mathrm{det}} = 2\eta_Z - \lambda_D$$

is the reserve of the two-exponent transfer, provided the pair is certified.

Proposition 3.2 (Three determinant statuses)**.** *For a certified pair:*

(i) if $\rho_{\mathrm{det}} > 0$*, the flatness estimate has a fixed power reserve;*

(ii) if $\rho_{\mathrm{det}} = 0$*, it reaches the threshold with no reserve;*

(iii) if $\rho_{\mathrm{det}} < 0$*, the recorded pair fails this certificate, but does not prove the target false.*

Proof. Read the exponent in corollary 2.2. ☐

If only $\eta_Z = 0$ is available, compatibility forces $\lambda_D = 0$. Thus a positive determinant-energy loss requires a compensating positive zero-mode gain in this ledger.

4 Independence from the physical endpoint

The complete route also has a synthesis operator whose separately derived fixed-power cost is denoted by Λ_{phys}. The inherited rule is

$$\boxed{\Lambda_{\mathrm{phys}} < \frac{1}{400}.}$$

Equality is a stop [1, 7, 8].

Theorem 4.1 (No exchange rate between ledgers)**.** *Without an additional literal intertwining theorem, neither* ρ_{det} *nor the pair* (λ_D, η_Z) *imposes any bound on* Λ_{phys}*. Conversely, the value of* Λ_{phys} *imposes no determinant-energy or zero-mode estimate. Therefore the audited route requires the conjunction*

$$\boxed{\lambda_D \le 2\eta_Z \quad \textit{and} \quad \Lambda_{\mathrm{phys}} < \frac{1}{400},}$$

not a merged inequality such as $\Lambda_{\mathrm{phys}} < 1/400 + \rho_{\mathrm{det}}$*.*

Proof. Take any finite coefficient vector realizing prescribed nonzero statistics D and Z. On an independent two-dimensional synthesis space, let the physical comparison operator be $\mathrm{diag}(1, X^L)$, with arbitrary $L \ge 0$. Changing L changes its condition-number exponent but leaves D, Z, and their reserve unchanged. Conversely, fix the synthesis map to be an isometry and vary the coefficient vector between a constant vector and a zero-sum vector. This changes the determinant flatness without changing the physical operator norm.

These direct-product models do not describe the arithmetic coefficient. They prove the logical point: a relationship between the ledgers requires a new map-specific theorem and cannot be created by bookkeeping. ☐

Corollary 4.2 (Different equality rules)**.** *The boundary*

$$\lambda_D = 2\eta_Z$$

passes the determinant compatibility test with zero reserve, while

$$\Lambda_{\mathrm{phys}} = \frac{1}{400}$$

fails the strict physical test. The first equality may not be cited as slack for the second.

5 No-double-charging protocol

Definition 5.1 (Typed loss record)**.** A fixed-power record consists of a source identifier, an exponent, and exactly one role:

$$\texttt{energy}, \quad \texttt{zero}, \quad \texttt{physical}, \quad \texttt{soft remainder}.$$

If one analytic estimate genuinely enters two inequalities, the proof must display the joint inequality. Merely copying its exponent into two rows is duplicate charging.

Proposition 5.2 (Valid ledger composition)**.** *A proof-carrying compatibility certificate must:*

(i) cite distinct proved bounds for D_X and Z_X, or one explicit joint theorem;

(ii) compute ρ_{det} once;

(iii) list every physical map cost once under Λ_{phys};

(iv) reject any repeated source identifier unless an exact factorization shows two disjoint uses; and

(v) leave open exponents open rather than assigning them zero.

Proof. Items (i)–(iii) reproduce the inequalities used in corollary 2.2 and theorem 4.1. Repeating a multiplicative factor does not create a new estimate, proving (iv). Assigning zero to an open term asserts a subpower theorem that has not been proved, proving (v). □

6 Audit of TPC-109–111

TPC-109 proves a sharp phase-saturation theorem for the coherent terminal-prime square. It supplies neither a growing energy lower bound nor a zero-mode upper bound [2].

TPC-110 proves the exact determinant-binning SVD and factors

$$D_X = E_X \alpha_X \overline{m}_X.$$

It shows that support alone cannot lower-bound the kernel angle, but does not estimate that angle for the growing literal coefficient [3].

TPC-111 proves exact coarsening invariance and sharp signed-prefix duality. Its outer criterion would imply a zero-mode exponent if the literal signed discrepancies were bounded, but no such growing arithmetic bound is proved [4].

Theorem 6.1 (Current compatibility verdict)**.** *The established results through TPC-111 certify no pair (λ_D, η_Z) in the exponent cone. Therefore the determinant–zero route is currently*

$$\boxed{\textit{open inputs; compatibility not yet testable.}}$$

It is neither a passed gate nor a global impossibility theorem.

Proof. TPC-110 supplies no lower bound of the required form for D_X, and TPC-111 supplies no upper bound of the required form for Z_X. By the typed-certificate definition, there is no certified pair to insert into corollary 2.2. □

7 Stop/go table and claim boundary

Gate	Go certificate	Stop or reroute certificate
Energy	Literal lower bound with a recorded λ_D	Actual upper obstruction excluding the desired loss profile
Zero mode	Literal upper bound with a recorded η_Z	Unavoidable signed-prefix or outer residual above target
Compatibility	$\lambda_D \le 2\eta_Z$	Recorded pair has $\lambda_D > 2\eta_Z$; seek sharper input
Physical endpoint	$\Lambda_{\text{phys}} < 1/400$	$\Lambda_{\text{phys}} \ge 1/400$, including equality

What is proved. The exponent formula, compatibility cone, reserve classification, ledger independence, no-double-charging protocol, and current "inputs open" verdict are rigorous L0/L1 audit results.

What is not proved. Neither input exponent is established for the actual growing coefficient. No physical synthesis bound is established. No L2 fixed-h_0 saving, parity breach, Hardy–Littlewood asymptotic, prime-pair lower bound, or twin-prime theorem is claimed. Nothing is specialized to $h_0 = 2$.

References

[1] Liang Wang. TPC-MVP1: A Falsifiable Proof Architecture for Fixed-Shift Prime-Pair Correlations: Verified Interfaces, Explicit Working Hypotheses, and Stop–Go Tests at the Parity Frontier. https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-mvp1-fixed-shift-proof-architecture, 2026.

[2] Liang Wang. Coherent Prime-Square Blocks on Literal Residual Fibers: Exact Gram Geometry, Sharp Saturation, and a Route-Switch Certificate. https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-109-coherent-prime-square-saturation, 2026.

[3] Liang Wang. Post-Bin Determinant Energy on the Actual Support: Exact Binning SVD, Kernel Angle, and a Natural-Scale Obstruction. https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-110-post-bin-determinant-energy, 2026.

[4] Liang Wang. The Distinguished Determinant Zero Mode: Exact Coarsening Invariance, Signed-Prefix Duality, and a Cancellation Obstruction. https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-111-distinguished-zero-mode-cancellation, 2026.

[5] Liang Wang. Matched Prime–Möbius Cutoff Shells: Rank–Two Factorization, Determinant–Frequency Dispersion, and the Distinguished Zero–Frequency Gate. https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-32-matched-cutoff-frequency-gate, 2026.

[6] Liang Wang. A Fixed-Shift Arithmetic Transfer Gap: From Literal Completion Certificates to Zero-Frequency Flatness. https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-81-fixed-shift-arithmetic-transfer-gap, 2026.

[7] Liang Wang. A Ten-Paper Fixed-Shift Completion-to-Flatness Audit: Exact Certificate Closures, the Determinant-Energy Gate, and a Zero-Slack Endpoint Ledger. https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-82-ten-paper-completion-to-flatness-audit, 2026.

[8] Liang Wang. A Ten-Paper Mass–Cancellation–Cover Audit: Exact Closures, Independent Arithmetic Gates, and a Stop/Go Roadmap. https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-92-ten-paper-mass-cancellation-cover-audit, 2026.